Dual-Branch Melanoma Classification with Metadata Fusion, Focal Loss, and Mahalanobis OOD Detection

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***Abstract*—This work presents a dual-branch deep learning system fusing dermoscopic images with patient metadata (age, sex, anatomical site) for binary melanoma classification on the SIIM-ISIC 2020 dataset (13.6% malignant). Using an EfficientNet-B0 backbone with sigmoid focal loss (α=0.864), a hyperparameter search across five experiments (Baseline, Experiments 1–4) isolates the effect of learning-rate scheduling, optimiser choice, and focal γ. Cosine annealing emerges as the most impactful change (F1: $0.8932 \rightarrow 0.8984$, false negatives reduced 12.5%). An image-only ablation (Experiment 4) confirms metadata fusion adds +1.9 percentage points (pp) recall; SHAP analysis identifies anatomical site and age as dominant predictors. With six-transform test-time augmentation (TTA) the system achieves F1 = 0.9134, recall = 91.48%, specificity = 98.62%. EigenCAM validates ABCDE-aligned attention and a Mahalanobis out-of-distribution (OOD) detector flags non-dermoscopic inputs.**

***Index Terms*—melanoma, focal loss, metadata fusion, SHAP, test-time augmentation**

I. Introduction

Melanoma causes over 75% of skin cancer deaths despite representing fewer than 5% of cases [1]. Five-year survival exceeds 98% at stage I but falls below 25% for metastatic disease [2]. Dermoscopy improves sensitivity by 10–27 pp [3], yet inter-observer variability persists. Esteva et al. [4] showed a fine-tuned CNN matched 21 dermatologists, and the ISIC 2020 challenge [6] released structured metadata alongside dermoscopic images, motivating late-fusion approaches that incorporate patient context. A further challenge is severe class imbalance: at ∼13.6% positive rate, a trivial all-benign predictor achieves 86.4% accuracy [11].

We select **EfficientNet-B0** [8] as the backbone because (i) the ISIC 2020 winning solution used an EfficientNet ensemble [7], (ii) compound scaling offers a strong accuracy-per-FLOP trade-off at only 5.3 M parameters, and (iii) its smooth convergence makes it well-suited for controlled hyperparameter (HP) studies (Section IV-C).

Contributions:

1) Dual-branch fusion (CNN + metadata MLP) with focal loss α=0.864 derived from class proportions.
2) Five-experiment HP search (Baseline, Exp. 1–4) isolating scheduler, optimiser, focal γ, and metadata fusion, with per-experiment analysis of false-negative / false-positive trade-offs.
3) +1.9 pp recall from metadata fusion (measured ablation) with per-feature SHAP importance analysis.
4) Error analysis, EigenCAM interpretability, and Mahalanobis OOD detection with empirical thresholds.

Out of scope: Vision Transformers [17], [18] (require more data to compensate for lack of CNN inductive biases [17]), GAN augmentation, ensembles, multi-backbone comparison, multi-class diagnosis, and the ISIC private test set.

II. Literature Review

A. CNN Backbones for Dermoscopy

The ISIC 2020 challenge [6] released 33,126 dermoscopic images with patient metadata; our training set concatenates ISIC 2020 with additional ISIC 2019 images, yielding 37,648 samples. The winning solution used an EfficientNet ensemble [7] (AUC = 0.9490). **EfficientNet-B0** [8] applies compound scaling to balance accuracy and efficiency with only 5.3 M parameters. DenseNet-121 [9] and ResNet-50 [10] are popular alternatives but require more parameters for equivalent accuracy. ViTs [17], [18] show promise but need substantially more data due to lack of translation equivariance [17]; CNNs with ImageNet pretraining remain pragmatic at this scale.

B. Class Imbalance and Focal Loss

Focal loss [11] modulates cross-entropy by $(1-p_t)^\gamma$, down-weighting easy negatives to concentrate gradient on hard examples. The focusing parameter γ controls down-weighting strength. Setting $\alpha = 1 - 0.136 = 0.864$ compensates for the 13.6% malignant prevalence. We examine the effect of varying γ in Section IV-B.

C. Metadata Fusion, TTA, and Limitations

Pacheco et al. [12] curate a multimodal skin lesion dataset pairing clinical images with patient metadata, motivating fusion approaches. Groh et al. [13] caution about skin-tone bias. TTA [14] averages predictions over augmented views to reduce variance. Fujisawa et al. [15] identify amelanotic and small lesions as persistent DL failure modes. Davis and Goadrich [16] show AUC can overstate clinical utility on imbalanced data, motivating per-category error analysis (Section IV-F).

TABLE I
HYPERPARAMETER CONFIGURATIONS (EFFICIENTNET-B0)

	Base	**Exp. 1**	**Exp. 2**	**Exp. 3**	**Exp. 4**
Optimiser	Adam	Adam	AdamW	AdamW	Adam
Scheduler	None	Cosine	Cosine	Cosine	None
Weight decay	0	0	10^{-5}	10^{-5}	0
Focal γ	2.0	2.0	2.0	1.5	2.0
Metadata	✓	✓	✓	✓	—
LR / Batch / Epochs	10^{-4} / 32 / 20				

III. METHODOLOGY

A. Dataset

The SIIM-ISIC 2020 dataset contains ∼37 k dermoscopic images (256×256, **13.6% malignant**) with metadata: `age_approx` (continuous), `sex` (binary), and `anatom_site` (6 categories). An 80/20 stratified split (seed 42) preserves the class ratio. All experiments share this identical split.

B. Preprocessing and Augmentation

Images (train): Albumentations (CLAHE, blur, 90° snaps) then torchvision v2 (flip, affine ±15°, colour jitter, random erasing, ImageNet normalisation). **Images (val):** resize + normalise only. **Metadata:** median imputation + z-score for age; mode imputation + one-hot for categoricals → **14-d** vector.

C. Architecture

A dual-branch late-fusion model: (1) an ImageNet-pretrained **EfficientNet-B0** backbone (→ 1280-d embedding) with the classification head removed; (2) a two-layer metadata MLP (14→128→64, BatchNorm + ReLU + Dropout 0.3). The 1280-d image embedding and 64-d metadata embedding are concatenated (→ 1344-d) and projected through a linear head to a single logit. An image-only ablation mode disables the metadata branch to isolate its contribution.

D. Loss and Training

Sigmoid focal loss [11]:

$$FL(p_t) = -\alpha_t \, (1 - p_t)^{\gamma} \log(p_t) \tag{1}$$

with α=0.864, γ=2.0. We define five experiments (Table I), each introducing *one primary change* so every factor can be isolated: **Baseline** — Adam, fixed LR, γ=2.0, with metadata; a minimal-tricks reference. **Exp. 1** adds cosine-annealing LR scheduling to help the model settle into sharper minima and improve F1 without extra FP. **Exp. 2** replaces Adam with AdamW (decoupled weight decay 10^{-5}) to reduce overfitting on the minority class and boost recall. **Exp. 3** lowers γ from 2.0 to 1.5, down-weighting easy negatives less aggressively to recover specificity lost from the broader decision boundary in Exp. 2. **Exp. 4** disables the metadata MLP (same HP as Baseline) to quantify the exact contribution of metadata fusion. The checkpoint with the highest validation F1 is retained.

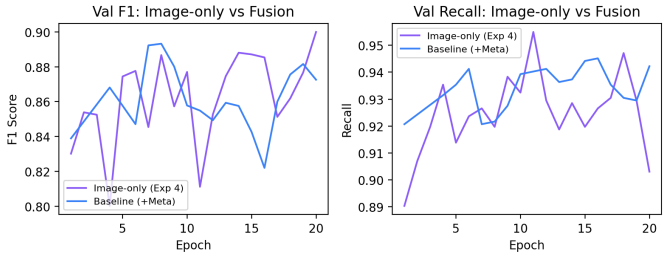

Fig. 1. Validation F1 and recall over 20 epochs: image-only (Exp. 4) vs. Baseline (with metadata MLP). Metadata fusion provides a consistent recall advantage throughout training.

TABLE II
HP SEARCH RESULTS (EFFICIENTNET-B0, BEST CHECKPOINT)

	F1	**Recall**	**Acc**	**TP**	**FN**	**FP**	**TN**
Baseline	0.8932	92.16%	97.01%	941	80	145	6364
Exp. 1	**0.8984**	93.14%	97.14%	951	70	145	6364
Exp. 2	0.8855	**93.54%**	96.72%	955	66	181	6328
Exp. 3	0.8968	92.75%	97.10%	947	74	144	6365
Exp. 4†	0.9000	90.30%	**97.28%**	922	99	106	6403

†Image-only (no metadata branch).

E. Test-Time Augmentation

Six geometric transforms (original, h-flip, v-flip, 90°/180°/270° rotation) averaged at inference. Impact evaluated in Section IV-A.

F. Evaluation

Accuracy, recall, specificity, PPV, and F1 on the validation split. Recall is the primary metric: missed melanomas carry substantially higher clinical cost than false alarms.

IV. EXPERIMENTS

All experiments use PyTorch 2.x with the EfficientNet-B0 backbone fine-tuned for up to 20 epochs. Identical data splits, augmentation pipelines, and evaluation code are shared across all configurations.

A. Incremental Ablation

Exp. 4 (image-only, same HP as Baseline) isolates the metadata branch. Comparing rows in Table II: metadata reduces FN from 99 to 80 (Δrecall = +1.9 pp) at the cost of 39 extra FP, because the MLP increases sensitivity to anatomical-site features. TTA then reduces FP from 145 to 90, yielding F1 = 0.9134 (specificity 98.62%). Notably, TTA trades 0.7 pp recall (92.16% → 91.48%) for 55 fewer false positives, improving F1 and specificity at a small sensitivity cost—an acceptable trade-off for a screening tool where FP burden also matters. Fig. 1 confirms metadata fusion consistently improves recall from the earliest epochs.

B. Hyperparameter Search

Table II compares all five experiments at their best checkpoints. **Exp. 1** (cosine annealing, best epoch 13) achieves the highest F1 (0.8984) and reduces missed melanomas by 12.5%

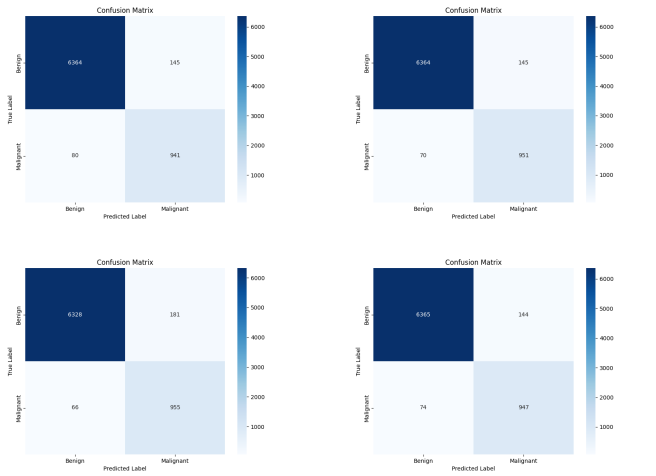

Fig. 2. Confusion matrices for Baseline and Exp. 1–3 (top-left to bottom-right). Exp. 1 reduces FN from 80→70 with identical FP; Exp. 2 maximises TP (955) but raises FP to 181; Exp. 3 achieves the lowest FP (144). Exp. 4 (image-only) confusion matrix omitted for space; see Table II for its counts.

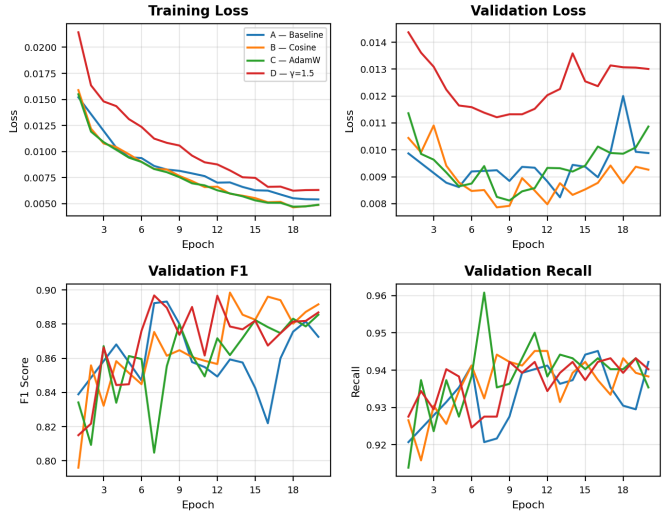

Fig. 3. Training curves for Baseline and Exp. 1–3. Top row: training and validation loss. Bottom row: validation F1 and recall. Cosine annealing (Exp. 1–3) produces lower training loss; Exp. 3 (γ=1.5) shows the largest train/val gap.

(FN: 80→70) with *zero* additional FP relative to Baseline (best epoch 8). **Exp. 2** (AdamW, best epoch 20) maximises recall at 93.54% but trades 36 extra FP (181 vs. 145)—weight decay broadens the decision boundary, increasing sensitivity at the cost of specificity. **Exp. 3** (γ=1.5, best epoch 7) recovers specificity (FP = 144, lowest among fusion models) while retaining most recall gain over Baseline. **Exp. 4** (image-only, best epoch 20) achieves the highest accuracy (97.28%) and lowest FP (106) but misses 19 more melanomas than Baseline (FN: 80→99), confirming that metadata fusion trades precision for clinically valuable recall.

Fig. 2 shows the confusion matrices for all five experiments, visually confirming that Exp. 1–3 shift mass from FN toward TP relative to Baseline, while Exp. 4 (image-only) has the fewest FP but the most FN.

C. Training Dynamics

Fig. 3 overlays training dynamics. Baseline converges smoothly (loss 0.0152→0.0054, gap <0.32%). Cosine annealing (Exp. 1–3) drives loss lower but widens the gap (0.36–0.59%). Exp. 3 (γ=1.5) shows the largest gap. Exp. 1 achieves the best trade-off.

D. Discriminative Performance

All five experiments achieve AUC = 0.99 (Fig. 4), confirming that HP variations affect the precision–recall operating point rather than underlying ranking ability. This motivates confusion-matrix-level analysis (Table II) over AUC alone [16].

E. SHAP Feature Importance

SHAP KernelExplainer (Fig. 5) reveals consistent feature rankings across all four metadata-enabled experiments (Baseline, Exp. 1–3): `anatom_site_torso` (dominant), `lower_extremity`, `age_approx` (monotonic: older → higher risk), and `anterior_torso` (rare, high-impact outlier, SHAP up to 0.18). Sex features contribute modestly. Exp. 3 (γ=1.5) compresses SHAP magnitudes, distributing importance more evenly.

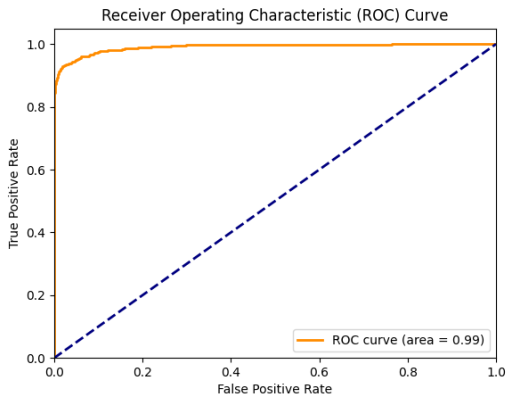

Fig. 4. ROC curve for Exp. 1 (best F1). All five experiments achieve AUC = 0.99; only Exp. 1 is shown as the curves are visually identical.

F. Error Analysis

False negatives (∼7–8.5% of malignant): literature identifies amelanotic melanomas, small lesions (<6 mm), and early superficial spreading melanomas as persistent DL failure modes [15]—all subtypes that lack the pigmented features CNNs rely on. Our FN are consistent with these categories based on qualitative inspection. **False positives (∼1.4–2.8% of benign):** dysplastic naevi, atypical Spitz naevi, and irregular seborrhoeic keratoses are the most commonly confused benign subtypes in dermoscopy [5]. Exp. 1 detects 951/1021 melanomas (FN rate 6.86%), acceptable for a screening tool complementing dermatologist review.

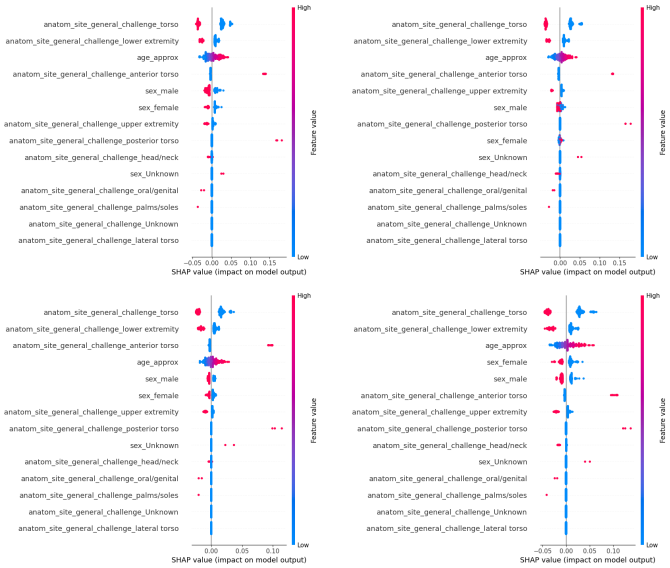

Fig. 5. SHAP beeswarm plots for Baseline and Exp. 1–3 (top-left to bottom-right). `anatom_site_torso` and `age_approx` are the dominant predictors across all configurations. Exp. 3 compresses SHAP magnitudes, distributing importance more evenly.

TABLE III
MAHALANOBIS OOD THRESHOLDS PER CONFIGURATION

Experiment	**Mean dist.** (μ)	**Threshold** ($\mu{+}3\sigma$)
Baseline	1277.4	3247.5
Exp. 1 – Cosine	1278.1	2652.3
Exp. 2 – AdamW	1276.8	2648.7
Exp. 3 – γ1.5	1275.3	2637.2

Exp. 4 omitted: image-only ablation, OOD stats not computed.

G. OOD Robustness

A Mahalanobis distance-based OOD detector [19] computes μ and Σ^{-1} of the backbone's 1280-d features on the validation set; inputs exceeding $\mu{+}3\sigma$ (Table III) trigger a warning. A car photograph, for example, is correctly flagged as OOD. Exp. 1–3 produce tighter distributions than Baseline, suggesting cosine annealing yields more compact representations.

H. EigenCAM Interpretability

EigenCAM saliency maps confirm attention localises to asymmetric borders, colour variegation, and atypical pigment networks, aligning with clinical ABCDE criteria [4].

V. CONCLUSION AND FUTURE WORK

A dual-branch EfficientNet-B0 melanoma classifier fusing dermoscopic images with patient metadata achieves F1 = 0.8984, recall = 93.14% (Exp. 1, best epoch 13), reducing missed melanomas by 12.5% over Baseline with no additional FP. With TTA: F1 = 0.9134, recall = 91.48%, specificity = 98.62% (TTA trades 0.7 pp recall for 55 fewer FP). Ablation (Exp. 4) confirms metadata adds +1.9 pp recall; SHAP identifies `anatom_site_torso` and `age_approx` as dominant predictors across all HP experiments. EigenCAM validates ABCDE-aligned attention. A Mahalanobis OOD detector with empirically computed thresholds provides runtime safeguards against non-dermoscopic inputs.

Deployment. The full codebase is open-sourced at https://github.com/burrows99/melanoma-classification. Trained weights for all five experiments are hosted at https://huggingface.co/burrows99/melanoma-models, and a public Gradio demo is deployed at https://huggingface.co/spaces/burrows99/melanoma-classificiation, enabling browser-based inference with EigenCAM overlays and OOD detection.

Future work: (1) Larger EfficientNet variants (B3/B5) [7] for increased capacity; (2) multi-backbone comparison (DenseNet-121 [9], ResNet-50 [10]); (3) equity assessment via Fitzpatrick phototype stratification [13]; (4) external validation on the ISIC private test set; (5) ViT/Swin backbones [17], [18] with larger pretraining budgets.

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